

Assignment 8 Report

Natural Language Processing Pipeline and Text Classification

Smart Home Device Support Message Classification

Item	Details
Course	AI Programming / Programming and Data Processing
Dataset	NLP_Dataset_16_Smart_Home_Device_Support.xlsx
Text Column	SupportMessage
Target Variable	SupportCategory
Feature Extraction	TF-IDF with unigrams and bigrams
Model Used	Logistic Regression
Prepared by	Vettivelu Prabaseelan
Submission Date	June 2026

Project Purpose

This report presents a complete NLP pipeline that converts raw smart home support messages into cleaned text, numerical TF-IDF features, and classification predictions. The business goal is to support faster and more accurate routing of customer support messages to the correct HomeIQ support team.

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1. Executive Summary

This report summarizes the Assignment 8 Natural Language Processing project using the Smart Home Device Support dataset. The project demonstrates how raw customer support messages can be cleaned, transformed into numerical features, and used to classify messages into business-relevant support categories.

Table 1: Executive summary of model pipeline and results

Project Element	Result
Dataset size	120 support messages
Number of features/columns	9 columns
Target classes	6 balanced support categories
Train/test split	96 training records and 24 testing records
Text representation	TF-IDF using unigrams and bigrams
Classifier	Logistic Regression
Test accuracy	100.00% on the test set
Important diagnostic finding	75 duplicated cleaned-text messages and strong category-specific anchor terms

Key Conclusion

The model achieved perfect test accuracy in the classroom dataset; however, the diagnostic review shows that this result is mainly explained by synthetic, templated text with strong anchor words and repeated cleaned messages. Therefore, the notebook proves that the NLP pipeline works, but it should not be interpreted as a deployment-ready system.

2. Business Problem and Project Objective

Smart home device companies receive many customer support messages through email, chat, web forms, and phone transcripts. These messages may concern Wi-Fi connectivity, device installation, security alerts, warranty returns, energy-saving features, or voice assistant issues. Manual classification can delay support routing and increase workload for service teams.

The business problem is to classify each HomeIQ customer support message into the correct support category. The model can support faster routing by directing security-alert messages to the appropriate monitoring team, warranty-return messages to returns support, and setup or connectivity issues to technical support.

The specific business objectives of the NLP pipeline are to:

- Reduce manual sorting of support messages.
- Route urgent security-alert messages more quickly.
- Identify recurring product-support themes such as warranty, connectivity, and energy-saving issues.
- Support data-driven improvement of help articles, troubleshooting workflows, and customer support staffing.

3. Dataset Overview

The dataset contains 120 records and 9 columns. Each row represents one customer support message related to smart home device support.

Table 2: Dataset overview

Item	Value
Rows	120
Columns	9
Missing values	0 in all columns
Duplicate full records	0
Text column	SupportMessage
Target variable	SupportCategory
Support categories	Connectivity, Energy_Saving, Installation, Security_Alert, Voice_Assistant, Warranty_Return

Note: Full-row duplicate records are zero; however, Section 10 separately reviews repeated cleaned-text patterns created after NLP preprocessing. This distinction is important because whole-row duplicates and duplicated cleaned messages measure different issues.

Table 3: Dataset columns and business meaning

Column	Business Meaning
SupportID	Unique identifier for each support record.
SupportDate	Date of the support message or request.
Channel	Customer contact channel such as email, chat, web form, or phone transcript.
City	Customer city; useful for geographic issue monitoring.
DeviceType	Smart home device involved in the message.
CustomerSegment	Customer segment, such as homeowner or renter.
SupportMessage	Raw text written or spoken by the customer; used as the main NLP input.
SupportCategory	Target label that the model predicts.
IssueSeverity	Severity indicator that can support prioritization.

Table 4: Target variable distribution

SupportCategory	Count	Percentage
Voice_Assistant	20	16.67
Security_Alert	20	16.67
Warranty_Return	20	16.67
Connectivity	20	16.67
Energy_Saving	20	16.67
Installation	20	16.67

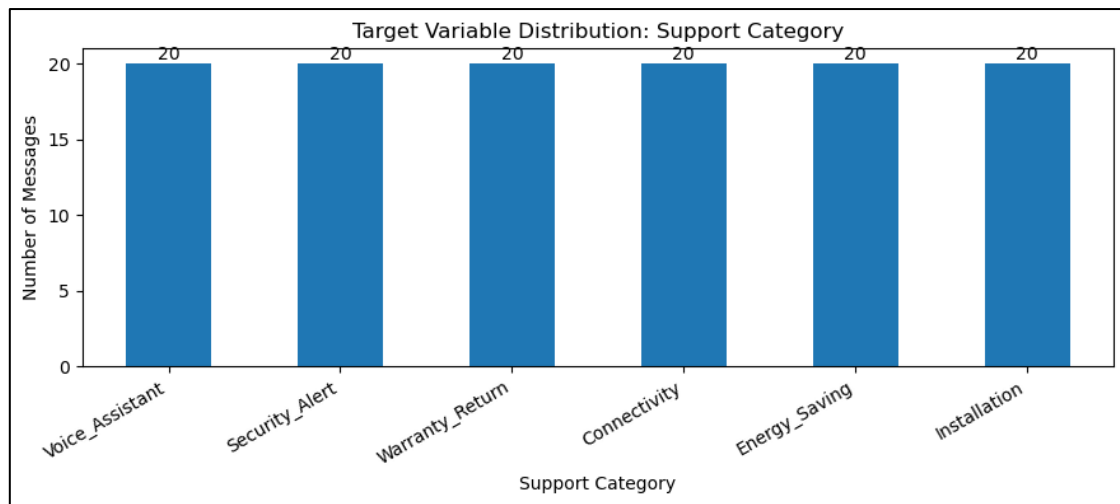


Figure 1: Target variable distribution across support categories

The target classes are balanced: each of the six support categories has 20 records, representing 16.67% of the dataset. This balance helps the model learn each class equally, but the small dataset size remains an important limitation.

4. Text Preprocessing

The main text column is SupportMessage. The notebook applies a standard NLP preprocessing pipeline to make the raw text more consistent before modelling.

Table 5: Text preprocessing steps and purpose

Step	Purpose
Lowercase conversion	Makes words such as “Alert” and “alert” consistent.
Punctuation removal	Removes symbols that do not add core meaning for this classification task.
Tokenization	Splits each message into individual words.
Stopword removal	Removes frequent low-information words such as “the”, “did”, and “when”.
Stemming	Reduces related words to root-like forms such as “working” to “work” and “warranty” to “warranti”.
Cleaned text creation	Combines the processed tokens into a cleaned text column for analysis and modelling.

The preprocessing examples show how raw support messages become more compact and model-ready. For example, in the security-alert message, stopwords such as “the”, “did”, “not”, “me”, and “when” are removed, while important words such as “app”, “notify”, “back”, “door”, and “opened” remain. After stemming, “notify” becomes “notifi” and “opened” becomes “open”.

Table 6: Side-by-side preprocessing examples from the dataset

Original_Text	Tokens	Tokens_After_Stopword_Removal	Stemmed_Tokens	Remov_Tokens	Cleane_Text
Voice commands worked yesterday, but today the...	[voice, commands, worked, yesterday, but, toda...	[voice, commands, worked, yesterday, today, sp...	[voic, command, work, yesterday, today, speake...	[but, the]	voic command work yesterday today speaker say ...
The app did not notify me when the back door o...	[the, app, did, not, notify, me, when, the, ba...	[app, notify, back, door, opened]	[app, notifi, back, door, open]	[the, did, not, me, when, the]	app notifi back door open
The light works near the router but fails in t...	[the, light, works, near, the, router, but, fa...	[light, works, near, router, fails, bedroom, b...	[light, work, near, router, fail, bedroom, bra...	[the, the, but, in, the, in]	light work near router fail bedroom brampton h...

Note: Some values in Table 6 are shortened with ellipses for readability in the report. The full preprocessing output is available in the submitted notebook.

5. Exploratory Text Analysis

Exploratory text analysis was used to identify the most frequent words in the cleaned support messages. These words reveal common product-support themes and help explain why the classifier performs strongly.

Table 7: Top 15 most common cleaned words

Word	Frequency
homeiq	48
work	25
camera	25
thermostat	20
say	15
app	15
light	15
hub	15
cannot	15
secur	15
alert	15
speaker	10
devic	10
door	10
smart	10

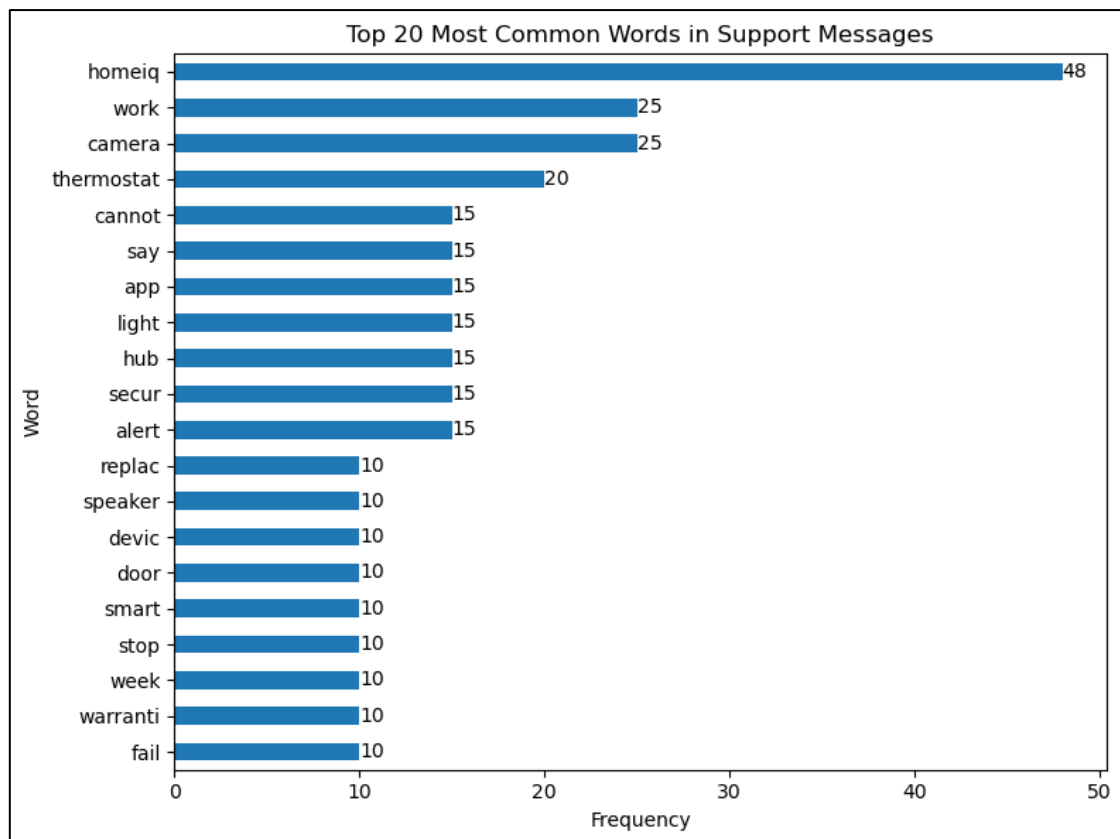


Figure 2: Top 20 most common words in support messages

The most frequent words include homeiq, work, camera, thermostat, app, light, hub, security, alert, warranty, and replacement-related terms. These terms show that customers contact HomeIQ about a limited number of recurring support themes, such as device failures, app notifications, security events, thermostat use, and warranty replacement requests.

6. POS Tagging and Named Entity Recognition

Three example records were selected from different support situations to demonstrate POS tagging and named entity recognition. The notebook confirmed that the required NLTK POS tagging and NER resources were available in the run, so NLTK POS tagging and NLTK NER were used. A rule-based business entity detector remains as a backup and supplement for environments where NLTK resources are unavailable.

Table 8: Selected records for POS tagging and NER

SupportID	City	DeviceType	SupportMessage	SupportCategory
SMT-005	Brampton	Smart Light	The light works near the router but fails in t...	Connectivity
SMT-002	Ottawa	Smart Speaker	The app did not notify me when the back door o...	Security_Alert
SMT-003	Mississauga	Hub	My smart light stopped working after two weeks...	Warranty_Return

Table 9: Sample POS tagging results

SupportID	Token	POS_Tag	Simplified_POS	SupportCategory
SMT-005	The	DT	Other	Connectivity
SMT-005	light	JJ	Adjective	Connectivity
SMT-005	works	NNS	Noun	Connectivity
SMT-005	near	IN	Other	Connectivity
SMT-005	the	DT	Other	Connectivity
SMT-005	router	NN	Noun	Connectivity
SMT-005	but	CC	Other	Connectivity
SMT-005	fails	VBZ	Verb	Connectivity
SMT-005	in	IN	Other	Connectivity
SMT-005	the	DT	Other	Connectivity
SMT-005	bedroom	NN	Noun	Connectivity
SMT-005	in	IN	Other	Connectivity
SMT-005	Brampton	NNP	Noun	Connectivity
SMT-005	HomeIQ	NNP	Noun	Connectivity
SMT-002	The	DT	Other	Security_Alert
SMT-002	app	NN	Noun	Security_Alert
SMT-002	did	VBD	Verb	Security_Alert
SMT-002	not	RB	Adverb	Security_Alert

Table 10: Named entity and business entity results

SupportID	Entity	Entity_Type	NER_Method	SupportCategory
SMT-005	Brampton	GPE	NLTK ne_chunk	Connectivity
SMT-002	No clear named entity detected		NLTK ne_chunk	Security_Alert
SMT-003	Smart Light	Device/Product	Rule-based fallback	Warranty_Return
SMT-003	week	Time/Date Expression	Rule-based fallback	Warranty_Return
SMT-003	weeks	Time/Date Expression	Rule-based fallback	Warranty_Return

POS tagging identifies the grammatical function of words. In the connectivity example, words such as light, router, and bedroom represent objects or locations, while verbs such as works and fails describe the problem. In the security-alert example, app and door identify the objects involved, while notify and opened describe the failed event. In the warranty-return example (SMT-003), words such as smart light, weeks, warranty, and replacement point to the product and return context, while stopped and working

describe the device failure that led to the warranty request. Entity detection can support business analysis by highlighting device names, locations, brands, platforms, or time expressions that may help route and prioritize messages.

7. Feature Extraction

Machine learning models cannot directly understand raw words. Therefore, the cleaned messages were converted into numerical features using TF-IDF. TF-IDF gives higher weight to words and phrases that are important in a message but not too common across every message.

Table 11: TF-IDF feature extraction summary

Item	Value
Feature extraction method	TF-IDF
N-gram setting	Unigrams and bigrams
TF-IDF matrix shape	120 rows x 276 features
Number of TF-IDF features	276
Reason for use	Works well for text classification by converting cleaned messages into sparse numerical features.

Table 12: Sample TF-IDF features

TF_IDF_Feature
alert
alert delay
alert june
alert one
alexa
alexa turn
almost
almost minut
anymor
app
app notifi
app recogn
app say
approv
approv warranti

The TF-IDF features include words and two-word phrases such as alert, alert delay, alexa, app notify, and other support-related patterns. These features allow the classifier to learn which terms are associated with each support category.

8. Text Classification Model

A Logistic Regression classifier was trained using the TF-IDF features. Logistic Regression is a suitable model because it performs well with sparse text features, supports multi-class classification, trains quickly on small datasets, and provides strong baseline performance for text classification tasks.

Table 13: Model training setup

Model Component	Description
X variable	Cleaned support-message text
y variable	SupportCategory
Train/test split	80/20 stratified split with random_state=42
Training records	96
Testing records	24
Model	Logistic Regression
Rationale	Chosen over tree-based models because sparse TF-IDF features are usually well-suited to linear text classifiers.

Table 14: Sample actual-versus-predicted results

Original_Text	Cleaned_Text	Actual_Category	Predicted_Category	Correct_Prediction	Predicted_Probability
The app did not notify me when the back door o...	app notifi back door open	Security_Alert	Security_Alert	Correct	0.5883
I want to rename the camera so Siri can recogn...	want renam camera siri recogn	Voice_Assistant	Voice_Assistant	Correct	0.509
The door sensor sent a motion alert at 2 AM, b...	door sensor sent motion alert one	Security_Alert	Security_Alert	Correct	0.5591
My smart light stopped working after two weeks...	smart light stop work two week need warranti r...	Warranty_Return	Warranty_Return	Correct	0.5416
The setup guide was clear, and I installed the...	setup guid clear instal camera oshawa minut	Installation	Installation	Correct	0.4753
Bluetooth pairing fails after the latest firmw...	bluetooth pair fail latest firmwar updat	Connectivity	Connectivity	Correct	0.6029
The thermostat report helped me lower energy u...	thermostat report help lower energi use week	Energy_Saving	Energy_Saving	Correct	0.5379
THE SECURITY CAMERA DISCONNECTS FROM WI-FI EVE...	secur camera disconnect wi fi everi night pm h...	Connectivity	Connectivity	Correct	0.4909

The sample prediction table shows that the model correctly classified the displayed test messages. The predicted probability column provides a simple indication of model confidence for each prediction.

9. Model Evaluation

The model was evaluated using accuracy, confusion matrix, and classification report metrics. These measures show both the overall performance and the class-level performance for each support category.

Table 15: Main evaluation metrics

Metric	Result	Business Interpretation
Accuracy	100.00%	All 24 test messages were classified correctly.
Precision	1.00 for every class	Every message predicted into a category was actually from that category.
Recall	1.00 for every class	Every actual category was fully identified in the test set.
F1-score	1.00 for every class	Balanced precision and recall were perfect for this test sample.

Table 16: Class-level evaluation summary

SupportCategory	precision	recall	f1-score	support
Connectivity	1.0	1.0	1.0	4.0
Energy_Saving	1.0	1.0	1.0	4.0
Installation	1.0	1.0	1.0	4.0
Security_Alert	1.0	1.0	1.0	4.0
Voice_Assistant	1.0	1.0	1.0	4.0
Warranty_Return	1.0	1.0	1.0	4.0

Table 17: Confusion matrix values

Actual Class	Predicted Connectivity	Predicted Energy_Saving	Predicted Installation	Predicted Security_Alert	Predicted Voice_Assistant	Predicted Warranty_Return
Actual Connectivity	4	0	0	0	0	0
Actual Energy_Saving	0	4	0	0	0	0
Actual Installation	0	0	4	0	0	0
Actual Security_Alert	0	0	0	4	0	0
Actual Voice_Assistant	0	0	0	0	4	0
Actual Warranty_Return	0	0	0	0	0	4

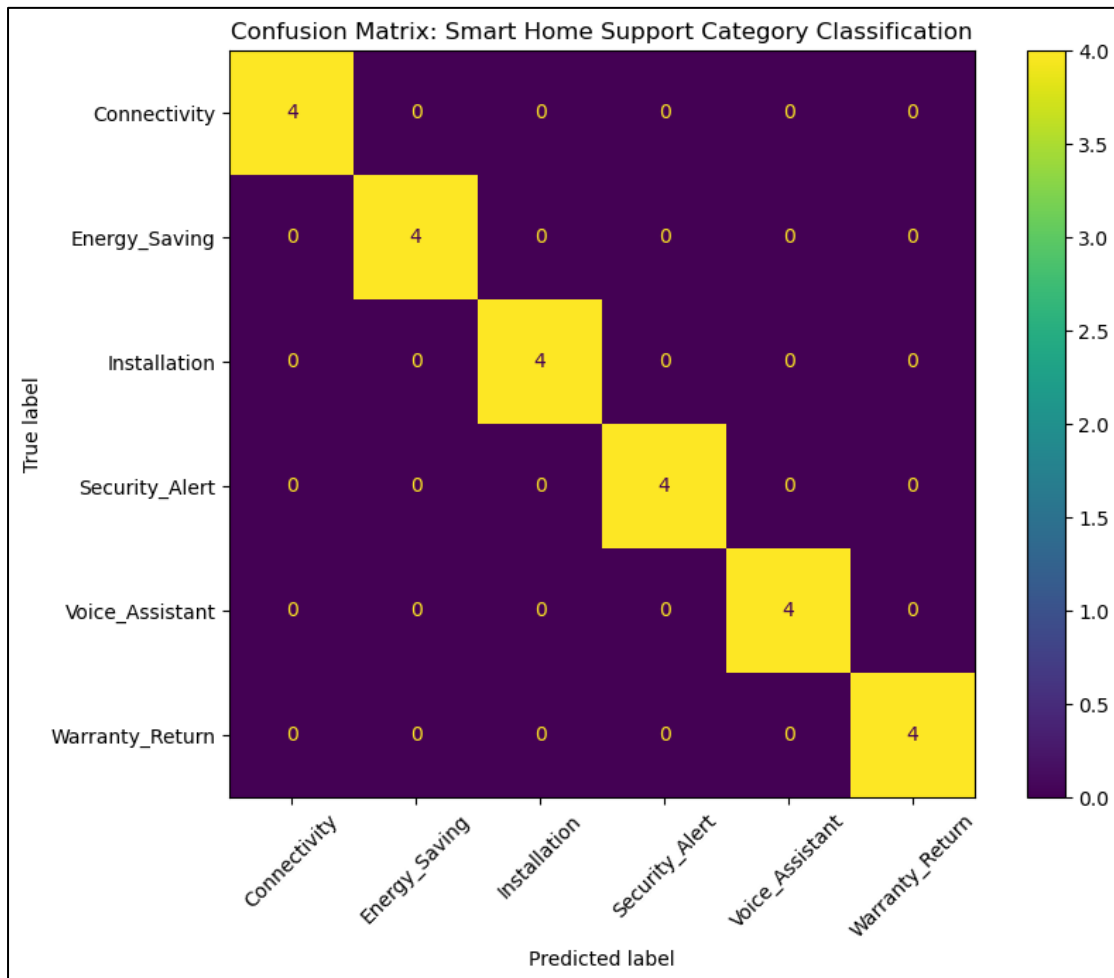


Figure 3: Confusion matrix visualization

The confusion matrix has all values on the diagonal and no off-diagonal mistakes. This means there were no misclassifications in the 24-message test set. However, this perfect result requires diagnostic review because the dataset is small, balanced, and synthetic.

10. Diagnostic Review of 100% Accuracy

A dedicated diagnostic check was added to investigate why the model reached 100% accuracy. This is important because perfect accuracy on a six-class text classification problem can indicate an artificially easy dataset or repeated text patterns.

Table 18: Diagnostic summary of perfect accuracy

Diagnostic Item	Result
Duplicated cleaned-text messages	75 duplicated cleaned-text messages
Percentage of all records	62.50% of the 120 records
Anchor-term pattern	Anchor terms appear almost exclusively in one category
Interpretation	The task is easier than real-world support classification because categories contain strong keyword signals.

Table 19: Examples of duplicated cleaned-text messages

SupportID	SupportCategory	Cleaned_Text
SMT-057	Voice_Assistant	alexa turn wrong light say live room lamp
SMT-076	Voice_Assistant	alexa turn wrong light say live room lamp
SMT-096	Voice_Assistant	alexa turn wrong light say live room lamp
SMT-103	Voice_Assistant	alexa turn wrong light say live room lamp
SMT-002	Security_Alert	app notifi back door open
SMT-053	Security_Alert	app notifi back door open
SMT-059	Security_Alert	app notifi back door open
SMT-099	Security_Alert	app notifi back door open
SMT-033	Energy_Saving	app say save percent chart hard understand
SMT-073	Energy_Saving	app say save percent chart hard understand
SMT-084	Energy_Saving	app say save percent chart hard understand
SMT-104	Energy_Saving	app say save percent chart hard understand

Table 20: Anchor-term counts by support category

Anchor Term	Connectivity	Energy_Saving	Installation	Security_Alert	Voice_Assistant	Warranty_Return
alert	0	0	0	15	0	0
assist	0	0	0	0	5	0
bluetooth	5	0	0	0	0	0
energi	0	10	0	0	0	0
instal	0	0	10	0	0	0
return	0	0	0	0	0	10
voic	0	0	0	0	5	0
warranti	0	0	0	0	0	10

Note: Anchor terms with zero counts across all categories were removed from this table. For example, Wi-Fi-related text may appear after preprocessing as separate tokens rather than the single token 'wifi'.

Critical Interpretation of Perfect Accuracy

The 100% accuracy should be understood as a prototype result under clean classroom-data conditions. The anchor-term table shows that terms such as alert, warranti, return, energi, instal, voic, and bluetooth are concentrated in specific categories. This means the model can classify the dataset using obvious category keywords rather than subtle language understanding.

11. Business Interpretation and Recommendation

The model predicts the support category for HomeIQ smart home customer messages. The text column used for modelling is SupportMessage, and the target variable is SupportCategory. The model classifies messages into Installation, Connectivity, Voice_Assistant, Energy_Saving, Security_Alert, and Warranty_Return.

The business value of this pipeline is that it demonstrates how NLP can route customer messages to the correct support team. In principle, such a system could reduce response time, prioritize urgent security-alert issues, and help managers understand common support themes.

These routing recommendations describe the target business workflow once the model is validated on larger real customer support data; they should not be read as proof that this prototype is already ready for automated deployment.

- Connectivity messages can be routed to network or device-pairing support.
- Security_Alert messages can be prioritized due to their safety-related nature.
- Warranty_Return messages can be routed to replacement or returns support.
- Energy_Saving messages can be connected to education, product guidance, or thermostat optimization support.
- Voice_Assistant messages can be routed to platform or integration support.

Recommendation

Use this model as a classroom prototype and proof of concept. Before real deployment, HomeIQ should retrain and evaluate the model using a larger sample of real support messages with more natural phrasing, spelling variation, mixed issues, and messages that do not contain obvious category keywords.

12. Limitations and Next Steps

Table 21: Limitations and recommended next steps

Limitation	Why It Matters	Recommended Next Step
Small dataset size	Only 120 records are available, with 24 test records.	Collect a larger real-world support dataset.
Synthetic and templated text	Messages are cleaner and more structured than real support text.	Validate using naturally written customer support messages.
Repeated cleaned-message patterns	Duplicates may allow the model to match familiar phrasing rather than generalize.	Remove duplicates or use grouped splitting to avoid train-test overlap.
Obvious category keywords	The model can rely on anchor words such as alert, warranty, energy, and return.	Test performance on messages without obvious category words.
Real-world language variation not fully represented	Real messages may include spelling errors, sarcasm, incomplete sentences, multiple issues, or indirect phrasing.	Monitor performance across channel, city, device type, and customer segment.

The most important next step is to validate the pipeline on a larger and more varied set of real support messages. Cross-validation, a larger held-out test set, and model comparison with Naive Bayes, Random Forest, or transformer-based methods would provide stronger evidence of business readiness.

Appendix A: Assignment Alignment Checklist

Table A1: Assignment requirement alignment checklist

Assignment Task	Completion in Report	Status
Task 1: Load and inspect the dataset	Dataset shape, columns, missing values, duplicate records, and target distribution are reported.	Completed
Task 2: Text preprocessing	Lowercasing, punctuation removal, tokenization, stopword removal, stemming, and cleaned text examples are explained.	Completed
Task 3: Exploratory text analysis	Most common words, word-frequency table, and bar chart are included and interpreted.	Completed
Task 4: POS tagging and NER	Three example records are analyzed using POS tagging and NER/entity detection.	Completed
Task 5: Feature extraction	TF-IDF feature extraction is explained with matrix shape and sample features.	Completed
Task 6: Text classification model	Logistic Regression model setup, train-test split, and predictions are summarized.	Completed
Task 7: Model evaluation	Accuracy, classification report, confusion matrix, and diagnostic review are included.	Completed
Task 8: Business interpretation	Business insight, limitation, and recommendation are clearly stated.	Completed